P.4 Inverse Functions

Name _____ Period ____ Date _____

AP Calculus P.4 · Student sheet

I can…

- decide whether a function is one-to-one.
- find and verify the inverse of a function.
- graph an inverse function and state its domain and range.

Vocabulary · fill in the missing word

One-to-one function — a function in which any two different inputs _____ to two different outputs

Horizontal-line Test — f is _____ if every horizontal line intersects its graph at most once

Inverse function — f^{-1}, where $f^{-1}(f(x)) = x$ and $f(f^{-1}(x)) = x$; the _____ of f^{-1} is the range of f

Part 1 · Undo it

$f(x) = x^3 + 1$. Fill in the table, then swap the columns.

x	−2	−1	0	1	2
f(x)					
Swapped: x					
Swapped: y					

Part 2 · Notes and examples (fill in each note, then work the example)

Horizontal-line Test: if every horizontal line intersects the graph of f at most _____, then f is one-to-one.

Example 1

Is each function one-to-one?

a. {(−3, 5), (−2, 9), (−1, 2), (0, 11), (1, −5)}

b. {(−2, 1), (−3, 2), (−10, 0), (1, 9), (2, 1)}

Graphs: the graph of f^{-1} is the reflection of the graph of f about the line _____.

Example 2

Find the inverse of {(−3, 5), (−2, 9), (−1, 2), (0, 11), (1, −5)} and give its domain and range.

Inverse: $f^{-1}(f(x)) = x$ for x in the domain of f, and $f(f^{-1}(x)) = x$ for x in the domain of _____.

Example 3

Show that $f(x) = 3x + 4$ and $g(x) = (\frac{1}{3})(x - 4)$ are inverses.

Finding f^{-1}: write $y = f(x)$, _____ x and y, and solve for y.

Example 4

Find the inverse of $f(x) = \frac{2x + 3}{x + 2}$ and give the domain and range of f and f^{-1}.

Example 5

Find the inverse of $f(x) = x^2 + 4$, $x \geq 0$.

Part 3 · Partner practice

Partner 1

Find the inverse of $f(x) = 4x + 2$.

Partner 2

Find the inverse of $f(x) = \sqrt[3]{(x + 10)}$.